

Week 10: Advanced Avellaneda-Stoikov, Structured Products, and Final Presentations

Extensions, the corporate view, and communicating results (Coincall guest session)

North Carolina State University | Summer 2026

NC STATE UNIVERSITY

Industry Mentor: Dendi Suhubdy (Bitwyre) · Guest: Fenni Kang (Coincall)

Learning Objectives

- ▶ Implement one advanced Avellaneda-Stoikov extension
- ▶ Understand how corporates use options via structured products
- ▶ Write and present a research-grade report

Lecture 1: Advanced Extensions

- ▶ Multi-asset market making across BTC and ETH options
- ▶ Adverse-selection-aware and signal-driven quoting
- ▶ Discrete-inventory formulations
- ▶ Robust optimization with a worst-case inventory penalty

Lecture 2 (Guest: Fenni Kang, Coincall): Structured Products

- ▶ Collars to fix a price band on a holding
- ▶ Accumulators / decumulators for staged accumulation or liquidation
- ▶ Coupon / yield-enhancement products
- ▶ How structured-product flow lands on the market-maker's book

Final Deliverables

- ▶ One-page extension proposal (day one)
- ▶ PyTorch implementation on the milestone infrastructure
- ▶ Quantitative comparison vs. the Week-9 baseline
- ▶ Final report (≤ 30 pages) and 20-minute presentation